

REGISTER ADDRESS

R=Read, W=Write

Address	Pre 3.0	3.0	3.1	5.14+	R	W	
0	x	x	x	x	*		Inputs, bits 0-15 [BBBBBBBTTxxxxxx] x=undef, T=tool, B=box
1	x	x	x	x	*	*	Outputs, bits 0-15 [BBBBBBBTTxxxxxx] x=undef, T=tool, B=box
2	x	x	x	x		*	SetOutputsBitsMask 0-15 [BBBBBBBTTxxxxxx] x=undef, T=tool, B=box
3	x	x	x	x		*	ClearOutputsBitsMask 0-15 [BBBBBBBTTxxxxxx] x=undef, T=tool, B=box
4	x	x	x	x	*		Analog input 0 (0-65535)
5	x	x	x	x	*	*	Analog input 0 domain e {0=current[mA], 1=voltage[mV]}
6	x	x	x	x	*		Analog input 1 (0-65535)
7	x	x	x	x	*	*	Analog input 1 domain e {0=current[mA], 1=voltage[mV]}
8	x	x	x	x	*		Analog input 2 (tool) (0-65535)
9	x	x	x	x	*	*	Analog input 2 (tool) domain e {0=current[mA], 1=voltage[mV]}
10	x	x	x	x	*		Analog input 3 (tool) (0-65535)
11	x	x	x	x	*	*	Analog input 3 (tool) range e {0=current[mA], 1=voltage[mV]}
16	x	x	x	x	*	*	Analog output 0 output (0-65535)
17	x	x	x	x	*	*	Analog output 0 output domain e {0=current[mA], 1=voltage[mV]}
18	x	x	x	x	*	*	Analog output 1 output (0-65535)
19	x	x	x	x	*	*	Analog output 1 output domain {0=current[mA], 1=voltage[mV]}
20	x	x	x	x	*	*	Tool output voltage (V) e {0V, 12V, 24V}
21	x	x	x	x	*		Tool digital input bits
22	x	x	x	x	*	*	Tool digital output bits
24	x	x	x	x	*		Euromap67 input bits (0-15)
25	x	x	x	x	*		Euromap67 input bits (16-32)
26	x	x	x	x	*		Euromap67 output bits (0-15) (read only!)
27	x	x	x	x	*		Euromap67 output bits (16-32) (read only!)
28	x	x	x	x	*		Euromap 24V voltage
29	x	x	x	x	*		Euromap 24V current
30	-	x	x	x	*		Configurable inputs, bits [BBBBBBBxxxxxxxx] x=undef, T=tool, B=box
31	-	x	x	x	*	*	Configurable outputs, bits [BBBBBBBxxxxxxxx] x=undef, T=tool, B=box
32	-	x	x	x		*	Bit mask configurable outputs, bits [BBBBBBBxxxxxxxx] x=undef, T=tool, B=box
33	-	x	x	x		*	Clear configurable outputs, bits [BBBBBBBxxxxxxxx] x=undef, T=tool, B=box
34-127	x	x	x	x			Reserved for future system variables
128-255	x	x	x	x	*	*	General purpose 16 bit registers
256-	x	x	x	x	*		Robot state
512-	x	x	x	x	*		Reserved
768-	x	x	x	x	*		Tool states
2048-	x	x	-	-	*		RT Machine control
Robot state							
256	x	x	x	x	*		Controller version high nubur
257	x	x	x	x	*		Controller version low number
258	x	x	x	x	*		Robot mode:
							CB3 and 3.1: Disconnected=0, Confirm_safety=1, Booting=2, Power_off=3, Power_on=4, Idle=5, Backdrive=6, Running=7
							CB2: No_controller=-1, Running=0, Freedrive=1, Ready=2, Initializing=3, Security_stoppe=4, Emergency_stopped=5, Fault=6, Not_connected=6
260	x	x	x	x	*		isPowerOnRobot
261	x	x	x	x	*		isSecurityStopped
262	x	x	x	x	*		isEmergencyStopped

263	x	x	x	x	*		isTeachButtonPressed
264	x	x	x	x	*		isPowerPuttonPressed
265	x	x	x	x	*		isSafetySignalSuchThatWeShouldStop
266				x	*		SafetyMode Refer to Primary Interface manual
270	x	x	x	x	*		Base joint angle (in mrad)
271	x	x	x	x	*		Shoulder joint angle (in mrad)
272	x	x	x	x	*		Elbow joint angle (in mrad)
273	x	x	x	x	*		Wrist1 joint angle (in mrad)
274	x	x	x	x	*		Wrist2 joint angle (in mrad)
275	x	x	x	x	*		Wrist3 joint angle (in mrad)
280	x	x	x	x	*		Base joint angle velocity (in mrad/s)
281	x	x	x	x	*		Shoulder joint angle velocity (in mrad/s)
282	x	x	x	x	*		Elbow joint angle velocity (in mrad/s)
283	x	x	x	x	*		Wrist1 joint angle velocity (in mrad/s)
284	x	x	x	x	*		Wrist2 joint angle velocity (in mrad/s)
285	x	x	x	x	*		Wrist3 joint angle velocity (in mrad/s)
290	x	x	x	x	*		Base joint current (in mA)
291	x	x	x	x	*		Shoulder joint current (in mA)
292	x	x	x	x	*		Elbow joint current (in mA)
293	x	x	x	x	*		Wrist1 joint current (in mA)
294	x	x	x	x	*		Wrist2 joint current (in mA)
295	x	x	x	x	*		Wrist3 joint current (in mA)
300	x	x	x	x	*		Base joint temperature (in C)
301	x	x	x	x	*		Shoulder joint temperature (in C)
302	x	x	x	x	*		Elbow joint temperature (in C)
303	x	x	x	x	*		Wrist1 joint temperature (in C)
304	x	x	x	x	*		Wrist2 joint temperature (in C)
305	x	x	x	x	*		Wrist3 joint temperature (in C)
Joint modes From version 1.7							
310	x	x	x	x	*		Base joint mode
311	x	x	x	x	*		Shoulder joint mode
312	x	x	x	x	*		Elbow joint mode
313	x	x	x	x	*		Wrist1 joint mode
314	x	x	x	x	*		Wrist2 joint mode
315	x	x	x	x	*		Wrist3 joint mode
							List of Joint Modes:
							JOINT_SHUTTING_DOWN_MODE = 236;
							JOINT_PART_D_CALIBRATION_MODE = 237;
							JOINT_BACKDRIVE_MODE = 238;
							JOINT_POWER_OFF_MODE = 239;
							JOINT_NOT_RESPONDING_MODE = 245;
							JOINT_MOTOR_INITIALISATION_MODE = 246;
							JOINT_BOOTING_MODE = 247;
							JOINT_PART_D_CALIBRATION_ERROR_MODE = 248;
							JOINT_BOOTLOADER_MODE = 249;
							JOINT_CALIBRATION_MODE = 250;

							JOINT_FAULT_MODE = 252;
							JOINT_RUNNING_MODE = 253;
							JOINT_IDLE_MODE = 255;
320	-	-	X	X	*		Base joint revolution count (number of full turns, typically 0 or 1)
321	-	-	X	X	*		Shoulder joint revolution count
322	-	-	X	X	*		Elbow joint revolution count
323	-	-	X	X	*		Wrist1 joint revolution count
324	-	-	X	X	*		Wrist2 joint revolution count
325	-	-	X	X	*		Wrist3 joint revolution count
TCP							
400	X	X	X	X	*		TCP-x in tenth of mm (in base frame)
401	X	X	X	X	*		TCP-y in tenth of mm (in base frame)
402	X	X	X	X	*		TCP-z in tenth of mm (in base frame)
403	X	X	X	X	*		TCP-rx in mrad (in base frame)
404	X	X	X	X	*		TCP-ry in mrad (in base frame)
405	X	X	X	X	*		TCP-rz in mrad (in base frame)
410	X	X	X	X	*		TCP-x speed in mm/s (in base frame)
411	X	X	X	X	*		TCP-y speed in mm/s (in base frame)
412	X	X	X	X	*		TCP-z speed in mm/s (in base frame)
413	X	X	X	X	*		TCP-rx speed in mrad/s (in base frame)
414	X	X	X	X	*		TCP-ry speed in mrad/s (in base frame)
415	X	X	X	X	*		TCP-rz speed in mrad/s (in base frame)
420	X	X	X	X	*		TCP-x offset in mm (in tool frame)
421	X	X	X	X	*		TCP-y offset in mm (in tool frame)
422	X	X	X	X	*		TCP-z offset in mm (in tool frame)
423	X	X	X	X	*		TCP-rx offset in mrad (in tool frame)
424	X	X	X	X	*		TCP-ry offset in mrad (in tool frame)
425	X	X	X	X	*		TCP-rz offset in mrad (in tool frame)
450	X	X	X	X	*		Robot current (in mA)
451	X	X	X	X	*		I/O current (in mA)
768	X	X	X	X	*		Tool Mode Refer to Primary Interface manual
769	X	X	X	X	*		Tool temperature (in C)
770	X	X	X	X	*		Tool current (in mA)
RTMachine							
2048	X	X	X	X		*	Split time (latches the actual time to the registers 2049-2053) calculates the time from last time the controller was restarted
2049	X	X	X	X	*		Milliseconds
2050	X	X	X	X	*		Seconds
2051	X	X	X	X	*		Minutes
2052	X	X	X	X	*		Hours
2053	X	X	X	X	*		Days

COIL ADDRESS						
Address	Pre 3.0	3.0	3.1	5.14+	R	W

0-15	x	x	x	x	*	
16-31	x	x	x	x	*	*
32-47	x	x	x	x		*
48-63	x	x	x	x		*
64-79	x	x	x	x	*	
80-95	x	x	x	x	*	
96-111	x	x	x	x	*	
112-127	x	x	x	x	*	
128-135	x	x	x	x	*	
136-143	x	x	x	x	*	*
144-151	x	x	x	x		*
152-159	x	x	x	x		*
260	x	x	x	x	*	
261	x	x	x	x	*	
262	x	x	x	x	*	
263	x	x	x	x	*	
264	x	x	x	x	*	
265	x	x	x	x	*	
300-427				x	*	*
500-627				x	*	

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Euromap67 input bits (0-15)

Euromap67 input bits (16-32)

Euromap67 output bits (0-15) (read only!)

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isPowerOnRobot

isProtectiveStopped

isEmergencyStopped

isTeachButtonPressed

isPowerPuttonPressed

isSafetySignalSuchThatWeShouldStop

128 General Purpose Input boolean registers (shared with RTDE boolean registers)

128 General Purpose Output boolean registers (shared with RTDE boolean registers)

